

PYTHON CORE PROFICIENCY EXAMINATION

Focus: Conditional Logic (if-elif-else), Mathematical Operators, & String Methods

Duration: 180 Minutes
Total Questions: 100 MCQs

Instructions: Evaluate each expression carefully. Assume standard Python 3.x behavior. Select the single best answer for each question.

SECTION A: CORE CONDITIONAL SYNTAX & OPERATORS (Q1 - Q25)

1. What will be the output of the following Python code snippet?

```
x = 15
if x % 2 == 0:
    print("Even")
else:
    print("Odd")
```

- A) Even
- B) Odd
- C) SyntaxError
- D) None

2. Which operator is utilized to verify if a number is completely divisible by another number without any remainder?

- A) /
- B) //
- C) %
- D) **

3. Consider the expression: `result = "Artificial Intelligence".startswith("Art")`. What is the data type of `result`?

- A) str
- B) int
- C) bool
- D) NoneType

4. What is the output of the following code structure?

```
age = 22
if age >= 18:
    print("Eligible", end=" ")
if age >= 21:
    print("Premium", end=" ")
else:
    print("Standard", end=" ")
```

- A) Eligible Premium
- B) Eligible Standard
- C) Premium
- D) Eligible

5. Python resolves `elif` as an abbreviation for which structural phrase?

- A) else line if
- B) else if
- C) either if
- D) execute if

6. What will be the output if the string method `endswith()` is applied as shown below?

```
filename = "report_v1.pdf"
if filename.endswith(".PDF"):
    print("Valid Document")
else:
    print("Invalid Format")
```

- A) Valid Document
- B) Invalid Format
- C) SyntaxError
- D) AttributeError

7. What is the value of result after executing this logic?

```
val = -7
if val > 0 or val % 2 != 0:
    result = "Match"
else:
    result = "No Match"
```

- A) Match
- B) No Match
- C) ValueError
- D) None

8. Evaluate the output of the following nested system:

```
score = 85
if score > 70:
    if score > 90:
        print("Grade A")
    else:
        print("Grade B")
else:
    print("Grade C")
```

- A) Grade A
- B) Grade B
- C) Grade C
- D) IndentationError

9. Which mathematical evaluation represents the correct calculation for an age threshold requirement?

```
birth_year = 2008
current_year = 2026
if current_year - birth_year >= 18:
    print("Adult")
else:
    print("Minor")
```

- A) Adult
- B) Minor
- C) 2018
- D) TypeError

10. What does the expression `5 ** 2 // 3` evaluate to inside a conditional check?

- A) 8
- B) 2
- C) 1
- D) 0

11. What is the visual/structural indicator used in Python to group statements within an `if` block?

- A) Curly braces `{}`
- B) Parentheses `()`
- C) Indentation (whitespace)
- D) Semicolons `;`

12. Review the behavior of the `startswith()` call with multiple parameters supplied as a tuple:

```
url = "https://example.com"
if url.startswith(("http://", "https://")):
    print("Secure or Web Protocol")
else:
    print("Local Path")
```

- A) Secure or Web Protocol
- B) Local Path
- C) TypeError
- D) NameError

13. What output will the following arithmetic-driven conditional loop or structure give?

```
num = 46
if num % 10 == 6:
    print("Ends with six")
else:
    print("Does not end with six")
```

- A) Does not end with six
- B) Ends with six
- C) 6
- D) SyntaxError

14. What occurs when an if condition statement lacks a trailing colon?

- A) The code runs but behaves unpredictably.
- B) A runtime ValueError is raised.
- C) A compile-time SyntaxError occurs.
- D) Python implicitly inserts the colon.

15. What will the following expression print?

```
s = "Data Science"
if not s.startswith("data"):
    print("Case Sensitive Verification")
else:
    print("Case Insensitive Verification")
```

- A) Case Sensitive Verification
- B) Case Insensitive Verification
- C) Nothing is printed
- D) AttributeError

16. Determine the final value assigned to discount:

```
age = 65
is_member = False
if age >= 60:
    if is_member:
        discount = 20
    else:
        discount = 10
else:
    discount = 0
```

- A) 20
- B) 10
- C) 0
- D) None

17. Which statement evaluates to True if num contains an odd integer?

- A) num / 2 == 1
- B) num % 2 != 0
- C) num // 2 == 0
- D) num % 2 == 0

18. What output does this block produce?

```
word = "Python"
if word.endswith("on") and len(word) > 5:
    print("Match Found")
else:
    print("Match Not Found")
```

- A) Match Found
- B) Match Not Found
- C) True
- D) False

19. How many execution branches can be processed concurrently inside a unified if-elif-else block?

- A) Exactly one branch.
- B) As many as evaluate to True.
- C) At least two branches.
- D) None, unless wrapped in a loop.

20. What is the value of output given this computation?

```
a, b = 12, 5
if a // b == 2:
    output = "Floor Division Correct"
else:
    output = "Normal Division Used"
```

- A) Floor Division Correct
- B) Normal Division Used
- C) 2.4
- D) TypeError

21. What happens when the condition expression evaluates to an integer value of 0?

```
if 0:
    print("True Branch")
else:
    print("False Branch")
```

- A) True Branch
- B) False Branch
- C) TypeError
- D) SyntaxError

22. What is printed by this string evaluation script?

```
text = "Developer"
if text.startswith("Dev") and text.endswith("per"):
    print("Full Verification Passed")
else:
    print("Verification Failed")
```

- A) Verification Failed
- B) Full Verification Passed
- C) True
- D) AttributeError

23. Consider an identity token format check: "ID-9942".startswith("ID-"). What does this evaluation yield?

- A) True
- B) False
- C) None
- D) "ID-"

24. What will be the console output of this code snippet?

```
num = 17
if num % 2 != 0 and num > 10:
    print("Condition Alpha")
elif num % 2 != 0:
    print("Condition Beta")
else:
    print("Condition Gamma")
```

- A) Condition Beta
- B) Condition Gamma
- C) Condition Alpha
- D) Alpha and Beta

25. What is the standard purpose of an else block in conditional routing structures?

- A) To check secondary condition logic if the main path fails.
- B) To guarantee execution when no prior explicit criteria match.
- C) To terminate execution loops instantly.
- D) To repeat previous logical expressions recursively.

SECTION B: STRING ALGORITHMS & SUBSTRING PARSING (Q26 - Q50)

26. Review the text extraction structure and decide what gets printed:

```
phrase = "The quick brown fox"
if phrase.endswith("fox "):
    print("Trailing Space Matched")
else:
    print("Strict Match Failed")
```

- A) Trailing Space Matched
- B) Strict Match Failed
- C) SyntaxError
- D) None of the above

27. What is the value of status after evaluating this expression?

```
msg = "Error code: 404"
if msg.startswith("Error"):
    status = "Fail"
elif msg.endswith("404"):
    status = "Not Found"
else:
    status = "Success"
```

- A) Fail
- B) Not Found
- C) Success
- D) Fail Not Found

28. What will the following expression evaluate to in Python?

```
s = ""
if s.startswith("") and s.endswith(""):
    print("Empty String Identity")
else:
    print("Null Error")
```

- A) Empty String Identity
- B) Null Error
- C) TypeError
- D) Nothing is printed

29. What is the resulting console log for this string slices test block?

```
email = "admin@company.org"
if email.endswith(".com"):
    print("Commercial Domain")
elif email.endswith(".org"):
    print("Organization Domain")
else:
    print("Alternative Domain")
```

- A) Commercial Domain
- B) Organization Domain
- C) Alternative Domain
- D) ValueError

30. If text = "Py", what does text.startswith("Python") output?

- A) True
- B) False
- C) IndexError
- D) ValueError

31. What is the structural flaw or output of the code shown below?

```
username = "guest_user"
if username.endswith("user"):
print("Access Allowed")
```

- A) Access Allowed
- B) SyntaxError
- C) IndentationError
- D) NameError

32. Evaluate this block that incorporates string slicing indices inside the startswith() method parameter constraints:

```
sentence = "The solution is simple"
if sentence.startswith("solution", 4):
    print("Offset Substring Found")
else:
    print("Offset Substring Missing")
```

- A) Offset Substring Found
- B) Offset Substring Missing
- C) TypeError
- D) IndexError

33. What is the value of result following this evaluation?

```
val = "12345"
if val.startswith("1") and int(val) % 2 != 0:
    result = "Odd Starter"
else:
    result = "Standard Check"
```

- A) Odd Starter
- B) Standard Check
- C) TypeError
- D) ValueError

34. What is the outcome of passing an integer parameter into startswith() inside an conditional block?

```
code = "2026_exam"
if code.startswith(2026):
    print("Current Year Exam")
```

- A) Current Year Exam
- B) False branch execution
- C) TypeError
- D) AttributeError

35. Evaluate this structure utilizing custom boundaries for string parsing:

```
data = "BetaVersion"
if data.endswith("Version", 0, 11):
    print("Boundaries Matched")
else:
    print("Boundaries Out of Range")
```

- A) Boundaries Matched
- B) Boundaries Out of Range
- C) IndexError
- D) TypeError

36. What output is yielded by the following script block?

```
device = "Smart Fridge"
if "Smart" in device and device.endswith("Fridge"):
    print("IoT Appliance Verified")
else:
    print("Legacy System")
```

- A) Legacy System
- B) IoT Appliance Verified
- C) SyntaxError
- D) TypeError

37. Determine the behavior of this code checking for empty parameters inside a tuple structure:

```
text = "Logistics"
if text.startswith(("", "Log")):
    print("Instant Positive")
else:
    print("Negative")
```

- A) Instant Positive
- B) Negative
- C) TypeError
- D) ValueError

38. What will the following expression print out?

```
word = "Mechanics"
if word.endswith(("ics", "ic")):
    print("Science/Field Classification")
else:
    print("General Term")
```

- A) Science/Field Classification
- B) General Term
- C) TypeError
- D) None

39. What will happen if you attempt to use a list instead of a tuple for multi-argument matching within `startswith()`?

```
name = "Dr. Smith"
if name.startswith(["Dr.", "Mr."]):
    print("Salutation Found")
```

- A) Salutation Found
- B) Code returns False silently.
- C) A `TypeError` is raised.
- D) An `AttributeError` is raised.

40. Analyze this case-transforming structural logic:

```
title = "python bootcamp"
if title.upper().startswith("PYTHON"):
    print("Bootcamp Match")
else:
    print("No Bootcamp Match")
```

- A) Bootcamp Match
- B) No Bootcamp Match
- C) `AttributeError`
- D) `SyntaxError`

41. What output is observed when parsing single space validation blocks?

```
space = " "
```

```
if space.startswith(" ") and space.endswith(" "):
    print("Whitespace Balanced")
else:
    print("Imbalanced")
```

- A) Whitespace Balanced
- B) Imbalanced
- C) Nothing is printed
- D) `TypeError`

42. Evaluate the output of the compound logical test configuration below:

```
token = "JWT_secret_key"
if token.startswith("JWT") and not token.endswith("public"):
    print("Valid Secret Key Asset")
else:
    print("Public/Unsecured Asset")
```

- A) Valid Secret Key Asset
- B) Public/Unsecured Asset
- C) SyntaxError
- D) None

43. Consider the execution of the following specific numeric string format tracking structure:

```
serial = "99-XYZ-2026"
if serial.endswith("2026") and int(serial[:2]) % 3 == 0:
    print("Validated Serial Entry")
else:
    print("Invalid Entry Standard")
```

- A) Validated Serial Entry
- B) Invalid Entry Standard
- C) ValueError
- D) TypeError

44. What output is thrown by the script when evaluating multi-line formatted structures?

```
document_heading = "\nHeader Content"
if document_heading.startswith("\n"):
    print("Newline Shift Identified")
else:
    print("Inline Document Flow")
```

- A) Newline Shift Identified
- B) Inline Document Flow
- C) EscapeError
- D) SyntaxError

45. What will the conditional evaluation of this slice assignment produce?

```
sample = "Microservices"
if sample[0:5].endswith("o"):
    print("Substring Check Complete")
else:
    print("Substring Check Mismatched")
```

- A) Substring Check Complete
- B) Substring Check Mismatched
- C) IndexError
- D) TypeError

46. What type of comparison expression is explicitly produced natively via string methods like `startswith()`?

- A) Identity evaluation expression.
- B) Lexicographical comparison operations.
- C) Boolean confirmation flags.
- D) RegEx pattern arrays.

47. Analyze the performance profile or output of this block:

```
label = "Architecture"
if label.startswith("Arch") or label.endswith("ture") and label == "":
    print("Complex True")
else:
    print("Complex False")
```

- A) Complex True
- B) Complex False
- C) SyntaxError
- D) NameError

48. If a script sets `file_ext = "archive.tar.gz"`, what is evaluated by `file_ext.endswith(".gz")`?

- A) True
- B) False
- C) None
- D) ValueError

49. What is output by this block executing an explicit logic inversion?

```
domain_name = "localhost"
if not domain_name.endswith(".com"):
    print("Internal Network Asset")
else:
    print("Public Network Routing")
```

- A) Internal Network Asset
- B) Public Network Routing
- C) None
- D) SyntaxError

50. Which expression cleanly checks if a string s starts with 'A' and ends with 'Z'?

- A) s.startswith("A") and s.endswith("Z")
- B) s.startswith("A" and "Z")
- C) s.endswith("Z").startswith("A")
- D) s.startswith("A") or s.endswith("Z")

SECTION C: PARITY VALIDATION & APPLIED MATHEMATICAL TRIGGERS (Q51 - Q75)

51. What does the following program route print out?

```
number = 100
if number % 4 == 0 and number % 5 == 0:
    print("Double Divisible Parity")
else:
    print("Standard Parity Structure")
```

- A) Standard Parity Structure
- B) Double Divisible Parity
- C) True
- D) TypeError

52. If val = 13, what statement correctly isolates its parity profile?

- A) if val // 2 == 6: print("Even")
- B) if val % 2 != 0: print("Odd")
- C) if val % 2 == 0: print("Odd")
- D) if val ** 2 == 169: print("Even")

53. What is the output of the following mathematical computation system?

```
result_value = 45 * 2 + 10 / 2
if result_value % 2 == 0:
    print("Even Balance")
else:
    print("Odd Balance Balance")
```

- A) Even Balance
- B) Odd Balance Balance
- C) 95.0
- D) TypeError

54. What is the execution result when float representations are passed to parity verification operations?

```
val = 4.0
if val % 2 == 0:
    print("Float Even Standard")
else:
    print("Float Odd Standard")
```

- A) Float Even Standard
- B) Float Odd Standard
- C) TypeError
- D) SyntaxError

55. Review this logic checking matrix parameters:

```
n = -12
if n < 0 and n % 2 == 0:
    print("Negative Even Number")
else:
    print("Alternative Range Check")
```

- A) Negative Even Number
- B) Alternative Range Check
- C) ValueError
- D) None

56. What does this evaluation order print?

```
calc = (15 - 5) * 3 ** 2
if calc > 50 or calc % 2 == 0:
    print("Condition Met Successful")
else:
    print("Condition Met Failed")
```

- A) Condition Met Successful
- B) Condition Met Failed
- C) 90
- D) SyntaxError

57. If x = 7 and y = 8, evaluate the following code block outcome:

```
if (x * y) % 2 == 0:
    print("Product is Even")
else:
    print("Product is Odd")
```

- A) Product is Even
- B) Product is Odd
- C) 56
- D) TypeError

58. Determine the console response for this complex numeric format assignment:

```
val = 27
if val % 3 == 0:
    if val % 2 == 0:
        print("Divisible by 6")
    else:
        print("Divisible by 3 but Odd")
else:
    print("Not divisible by 3")
```

- A) Divisible by 6
- B) Divisible by 3 but Odd
- C) Not divisible by 3
- D) IndentationError

59. What mathematical operation returns the integer quotient discarding the fractional component during conditional logic setup?

- A) /
- B) %
- C) //
- D) div()

60. Analyze this compound operation execution:

```
value = 11 ** 2 - 1
if value % 10 == 0 and value % 4 == 0:
    print("Double Criterion Verification")
else:
    print("Failed Compound Criterion")
```

- A) Double Criterion Verification
- B) Failed Compound Criterion
- C) 120
- D) ValueError

61. What will be the output of this conditional expression configuration?

```
count = 87
if count % 2 != 0:
    if count > 50:
        print("High Odd")
    else:
        print("Low Odd")
else:
    print("Even")
```

- A) High Odd
- B) Low Odd
- C) Even
- D) SyntaxError

62. Which expression correctly validates that a variable number is a multiple of both 3 and 7?

- A) `number % 3 == 0 or number % 7 == 0`
- B) `number % 21 == 0`
- C) `number % 3 == number % 7`
- D) `(number / 3) == 7`

63. Evaluate the following mathematical statement block:

```
x = 14
if x * 2 // 4 >= 7:
    print("Condition Met Alpha")
else:
    print("Condition Met Beta")
```

- A) Condition Met Alpha
- B) Condition Met Beta
- C) 7
- D) TypeError

64. Determine the resulting log for this operation system:

```
steps = 0
if steps == 0:
    print("Stationary Phase")
elif steps % 2 == 0:
    print("Even Progression")
else:
    print("Odd Progression")
```

- A) Stationary Phase
- B) Even Progression
- C) Odd Progression
- D) RuntimeError

65. What value represents truthiness when evaluating conditional expressions?

- A) Any non-zero numerical integer value.
- B) Only the integer constant 1.
- C) Empty string variables.
- D) None constant mappings.

66. Review this math and tracking configuration block:

```
points = 50
if points - 25 * 2 == 0:
    print("Zero Reset Activation")
else:
    print("Residual Value Present")
```

- A) Zero Reset Activation
- B) Residual Value Present
- C) SyntaxError
- D) 0

67. What will the following evaluation print out?

```
number = 19
if number % 2 != 0:
    number += 1
if number % 2 == 0:
    print("Transformed to Even")
```

- A) Transformed to Even
- B) No execution output
- C) 20
- D) UnboundLocalError

68. Evaluate this block featuring relational string parsing combined with a modulus test:

```
code_string = "882"
if int(code_string) % 2 == 0 and code_string.startswith("8"):
    print("Valid Code Format Pattern")
else:
    print("Invalid Code Format Pattern")
```

- A) Valid Code Format Pattern
- B) Invalid Code Format Pattern
- C) ValueError
- D) TypeError

69. What will the following expression evaluate to?

```
val = 144
if val ** 0.5 == 12:
    print("Square Root Verified")
else:
    print("Square Root Failed")
```

- A) Square Root Verified
- B) Square Root Failed
- C) 12.0
- D) TypeError

70. Review the performance of the following script block:

```
x = 10
y = 20
if x > 5 and y // x == 2:
    print("Mathematical Operations Validated")
else:
    print("Mathematical Operations Failed")
```

- A) Mathematical Operations Validated
- B) Mathematical Operations Failed
- C) 2
- D) ZeroDivisionError

71. What value gets printed by the code block execution below?

```
number = 77
if number % 11 == 0:
    if number % 2 == 0:
        print("Even Eleventh")
    else:
        print("Odd Eleventh")
else:
    print("Base Value Out of Bounds")
```

- A) Even Eleventh
- B) Odd Eleventh
- C) Base Value Out of Bounds
- D) TypeError

72. Determine the assignment state of flag given the expression below:

```
score = 33
if score * 3 > 100:
    flag = "High Profile"
elif score % 3 == 0:
    flag = "Factor Profile"
else:
    flag = "Default Profile"
```

- A) High Profile
- B) Factor Profile
- C) Default Profile
- D) None

73. What is the output of this conditional expression configuration?

```
x = 64
if x % 8 == 0 and not x % 3 == 0:
    print("Clean Multiple")
else:
    print("Mixed Multiple")
```

- A) Clean Multiple
- B) Mixed Multiple
- C) True
- D) SyntaxError

74. Analyze the behavioral routing logic sequence here:

```
val = 1
if val > 0:
    if val < 0:
        print("Paradox Step One")
    elif val == 1:
        print("Target Step Confirmed")
else:
    print("Final Step Execution")
```

- A) Paradox Step One
- B) Target Step Confirmed
- C) Final Step Execution
- D) IndentationError

75. Which operator handles exponential math calculations inside Python expressions?

- A) ^
- B) **
- C) *
- D) ^^

SECTION D: APPLIED AGE CALCULATION & CHRONOLOGICAL ALGORITHMS (Q76 - Q100)

76. Review this standard profile logic structure:

```
birth_year = 2000
target_year = 2026
calculated_age = target_year - birth_year
if calculated_age >= 25:
    print("Milestone Passed")
else:
    print("Milestone Pending")
```

- A) Milestone Passed
- B) Milestone Pending
- C) 26
- D) TypeError

77. What happens if an input age parameter is represented purely as a string during validation?

```
reported_age = "18"
if reported_age >= 18:
    print("Access Verified")
```

- A) Access Verified
- B) Code executes false block silently.
- C) A TypeError is raised due to unsupported comparison operators.
- D) Python implicitly converts the string variable.

78. Determine the console log for this dual-condition demographic test:

```
age = 16
has_permit = True
if age >= 16 and has_permit:
    print("Authorized Driver Classification")
else:
    print("Unauthorized Status Flag")
```

- A) Authorized Driver Classification
- B) Unauthorized Status Flag
- C) SyntaxError
- D) None

79. Evaluate this tier mapping structural process script:

```
age = 12
if age < 5:
    print("Infant Tier")
elif age < 13:
    print("Child Tier")
elif age < 20:
    print("Teen Tier")
else:
    print("Adult Tier")
```

- A) Infant Tier
- B) Child Tier
- C) Teen Tier
- D) Adult Tier

80. What is the execution path and output for this system configuration code?

```
birth_year = 2010
current_year = 2026
age = current_year - birth_year
if age % 2 == 0:
    print("Even-Numbered Age: " + str(age))
else:
    print("Odd-Numbered Age: " + str(age))
```

- A) Even-Numbered Age: 16
- B) Odd-Numbered Age: 16
- C) Even-Numbered Age: 14
- D) TypeError

81. Analyze this nested confirmation check for premium clients:

```
age = 72
if age >= 65:
    if age >= 70:
        print("Premium Senior Tier")
    else:
        print("Standard Senior Tier")
else:
    print("Standard Adult Tier")
```

- A) Standard Senior Tier
- B) Premium Senior Tier
- C) Standard Adult Tier
- D) IndentationError

82. If age = 45, what output is produced by the following structure?

```
if age > 40:
    print("Category A", end="-")
if age > 30:
    print("Category B", end="-")
else:
    print("Category C", end="-")
```

- A) Category A-
- B) Category A-Category B-
- C) Category B-
- D) Category A-Category C-

83. Determine the output of the following chronological code segment:

```
birth_year = 1998
current_year = 2026
age = current_year - birth_year
if str(age).endswith("8"):
    print("Age Matrix Alignment Positive")
else:
    print("Age Matrix Alignment Negative")
```

- A) Age Matrix Alignment Positive
- B) Age Matrix Alignment Negative
- C) TypeError
- D) AttributeError

84. What happens when an `elif` check runs after its matching `if` has already resolved to `True`?

- A) The `elif` block expression is completely bypassed.
- B) The `elif` statement is evaluated regardless.
- C) Python execution throws a secondary `RuntimeWarning`.
- D) The script performs a fallback logical loop.

85. Review the structural logic checking identity configurations:

```
age = 19
status_string = "Student"
if age < 20 and status_string.startswith("Stud"):
    print("Eligible for Academic Pricing Package")
else:
    print("Standard Commercial Pricing Rules Applied")
```

- A) Standard Commercial Pricing Rules Applied
- B) Eligible for Academic Pricing Package
- C) `TypeError`
- D) `ValueError`

86. Evaluate this inverted threshold condition sequence:

```
age = 30
if not age < 25:
    print("Upper Dynamic Range Validation")
else:
    print("Lower Dynamic Range Validation")
```

- A) Upper Dynamic Range Validation
- B) Lower Dynamic Range Validation
- C) None
- D) `SyntaxError`

87. What logic expression verifies if age falls squarely between 13 and 19 inclusive?

- A) `13 <= age <= 19`
- B) `age == 13 or 19`
- C) `13 < age < 19`
- D) `age in range(13, 19)`

88. What is the console output for the following structural script?

```
birth_year = 2022
current_year = 2026
age = current_year - birth_year
if age * 4 == 16:
    print("Target Age Verification Confirmed")
else:
    print("Target Age Verification Failed")
```

- A) Target Age Verification Confirmed
- B) Target Age Verification Failed
- C) 16
- D) TypeError

89. Review the compound conditions processing inside this script block:

```
age = 5
if age % 2 != 0 and age < 10:
    print("Condition Type Alpha")
elif age % 2 != 0:
    print("Condition Type Beta")
```

- A) Condition Type Beta
- B) Condition Type Alpha
- C) No output is generated
- D) SyntaxError

90. Determine the final value of result from this block:

```
age = 20
if age >= 18:
    result = "Adult"
if age >= 60:
    result = "Senior"
else:
    result = "General Category"
```

- A) Adult
- B) Senior
- C) General Category
- D) None

91. Review this math transformation tracking structure:

```
birth_year = 1980
current_year = 2026
age = current_year - birth_year
if age // 10 == 4:
    print("In Fourth Decade Group")
else:
    print("Alternative Decade Group")
```

- A) In Fourth Decade Group
- B) Alternative Decade Group
- C) 4
- D) TypeError

92. What output is observed when passing boolean criteria matching flags into this code segment?

```
age = 25
is_registered = True
if age > 21:
    if not is_registered:
        print("Registration Request Required")
    else:
        print("Account Clearance Approved")
```

- A) Registration Request Required
- B) Account Clearance Approved
- C) None
- D) SyntaxError

93. What is the execution behavior when evaluating negative integers for age variables?

```
age = -5
if age < 0:
    print("System Error: Negative Value")
else:
    print("Valid System Age Model")
```

- A) System Error: Negative Value
- B) Valid System Age Model
- C) ValueError
- D) SyntaxError

94. Review this script parsing string elements and numeric components concurrently:

```
input_data = "Age: 35"
if input_data.startswith("Age") and int(input_data[-2:]) % 5 == 0:
    print("Clean Dynamic Match")
else:
    print("Dynamic Match Failure")
```

- A) Clean Dynamic Match
- B) Dynamic Match Failure
- C) ValueError
- D) TypeError

95. What output will the following expression execute?

```
age = 65
if age > 60:
    print("Senior Benefits Active", end=" ")
if age == 65:
    print("Exact Milestone Active", end=" ")
```

- A) Senior Benefits Active
- B) Senior Benefits Active Exact Milestone Active
- C) Exact Milestone Active
- D) SyntaxError

96. What is the visual signature or token used to open execution blocks across all conditional levels in Python?

- A) A semicolon token (;)
- B) A colon token (:)
- C) An arrow operator (->)
- D) A periods configuration indicator (. . .)

97. Analyze the value tracking resolution code shown below:

```
current_year = 2026
birth_year = 2006
if (current_year - birth_year) * 2 == 40:
    print("Calculation Balanced")
else:
    print("Calculation Imbalanced")
```

- A) Calculation Balanced
- B) Calculation Imbalanced
- C) 40
- D) TypeError

98. What output does the code snippet generate?

```
age = 17
classification = "Minor" if age < 18 else "Adult"
print(classification)
```

- A) Minor
- B) Adult
- C) SyntaxError
- D) None

99. Review this complex identity validation system performance:

```
user_profile = "AgeGroup_Adult"
if user_profile.startswith("Age") and user_profile.endswith("Adult"):
    print("Profile Pattern Authenticated")
else:
    print("Profile Pattern Replaced")
```

- A) Profile Pattern Replaced
- B) Profile Pattern Authenticated
- C) True
- D) AttributeError

100. What is the fundamental behavior of an empty conditional statement containing only a pass keyword?

```
if True:
    pass
else:
    print("Else Execution")
```

- A)** It throws an IndentationError.
- B)** It breaks out of the file execution.
- C)** It executes successfully without performing any operational actions.
- D)** It causes the program to fall through to the else statement.

OFFICIAL EXAMINATION ANSWER KEY

The following table provides the single correct option entry for each of the 100 evaluation items contained within this examination booklet.

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	21	B	41	A	61	A	81	B
2	C	22	B	42	A	62	B	82	B
3	C	23	A	43	A	63	A	83	A
4	A	24	C	44	A	64	A	84	A
5	B	25	B	45	A	65	A	85	B
6	B	26	B	46	C	66	A	86	A
7	A	27	A	47	A	67	A	87	A
8	B	28	A	48	A	68	A	88	A
9	A	29	B	49	A	69	A	89	B
10	A	30	B	50	A	70	A	90	C
11	C	31	C	51	B	71	B	91	A
12	A	32	A	52	B	72	A	92	B
13	B	33	A	53	B	73	A	93	A
14	C	34	C	54	A	74	B	94	A
15	A	35	B	55	A	75	B	95	B
16	B	36	B	56	A	76	B	96	B
17	B	37	A	57	A	77	C	97	A
18	A	38	A	58	B	78	A	98	A
19	A	39	C	59	C	79	B	99	B
20	A	40	A	60	B	80	A	100	C